Infectious Disease Simulation Case Study

Part I: Explore the VaxImpact Simulator here: https://herdimmunitysimulation.base44.app/
Spend a few minutes freely exploring the simulator and writing down notes about the following

Part I:
It started with a friendly greeting. On Monday morning, Skyline High's student body president welcomed a group of students to the weekly health sciences club meeting. Within 24 hours, five students were home sick, two teachers had called out, and rumors about a viral outbreak had started circulating.

As part of the Emergency Student Public Health Task Force, you have been called in to assess the situation, conduct data modeling through a simulator, and present a response plan to the school board.

Part II: Hands-On Activity (Contact Transmission Demo)

Objective: Model how easily disease spreads through contact.

Instructions:
1. Open your activity kit.
2. Two student tables will find a material inside their kits that simulates contamination, approximately 40% of the class.
3. Over the next minute, stand up, walk around the room, and interact as you normally would, high five classmates, borrow a pencil, or share an object.
4. After the minute is up, return to your seat.
5. Inspect your hands and record your observations.

Observation Questions:
What do you notice on your hands?
What surprised you about the results?
What do you think this represents in a real-world outbreak?

Part III: Simulator Analysis: VaxImpact

Access the simulator: https://herdimmunitysimulation.base44.app

Objective: Use the VaxImpact simulator to examine how vaccination rates affect the spread of disease in a closed environment similar to a school.

Scenario Setup:
There are concerns that the outbreak may be related to a drop in vaccine rates. Use the simulator to model disease spread at different vaccination levels.

Set the following parameters:

Transmission Rate: 75%
Recovery Time: 14 days
Mortality Rate: 1%
Population Density: 85%
Vaccine Effectiveness: 95%

Run 3 trials at these vaccination rates:

- 65%
- 80%
- 95%

Record the following for each trial:

Table 1: Measles Cases with Vaccination Rates set to 65%

Trial	Maximum Sick People in a single day (Peak Infections)	Total infected	% Vaccinated sick	% Unvaccinated sick
1				
2				
3				
Average				

Table 2: Measles Cases with Vaccination Rates set to 80%

Trial	Maximum Sick People in a single day (Peak Infections)	Total infected	% Vaccinated sick	% Unvaccinated sick
1				
2				
3				
Average				

Table 3: Measles Cases with Vaccination Rates set to 95%

Trial	Maximum Sick People in a single day (Peak Infections)	Total infected	% Vaccinated sick	% Unvaccinated sick

1				
2				
3				
Average				

Discuss:

- How did vaccination rates change the spread?
- Why are unvaccinated students at greater risk, even with high community coverage?
- What does this tell you about herd immunity?
- How does the simulator relate to what happened in the contact demo?

Part IV: Team Response Challenge

Your team must now create a one-minute pitch to the school board answering:

1. What is your theory on how the outbreak started and spread?
2. What is your recommended action plan? (Include data from both the hands-on activity and the simulator.)
3. What would you say to students, staff, and families to reduce panic and misinformation?

Roles (choose one per person):

- Public Health Officer
- Principal
- PTA Representative
- Student Body President
- Local Clinic Physician

Prepare your answers, support your plan with evidence, and be ready to present your recommendations.

Part V: Debrief and Reflection

- What did your team do well?
- What was the hardest part of planning a response?
- What would you change if you had more time or information?

You have completed a simulation-based investigation of a school outbreak using both a hands-on demonstration and a digital simulator. Your work reflects how scientific reasoning and public health strategies are used in real decision-making.